

JDScience BTEC Level 3 Structure and Bonding Answer Sheet

Unit 1 A1 - atomic structure, electron configuration and bonding.

Mark scheme

Task	Answer / marking points
A	K-40: 19p,21n,19e; Ca-40:20p,20n,20e; C-12:6p,6n,6e; C-13:6p,7n,6e; S-32:16p,16n,16e; Al-27:13p,14n,13e.
B	C: 2,4 / 1s ² 2s ² 2p ² ; O: 2,6 / 1s ² 2s ² 2p ⁴ ; P: 2,8,5 / 1s ² 2s ² 2p ⁶ 3s ² 3p ³ ; Na: 2,8,1 / 1s ² 2s ² 2p ⁶ 3s ¹ ; Mg: 2,8,2 / 1s ² 2s ² 2p ⁶ 3s ² .
C	Mg loses two electrons to form Mg ²⁺ ; oxygen gains two electrons to form O ²⁻ ; oppositely charged ions attract strongly; charges balance 1:1 so formula MgO.
D	Ordinary covalent bond: each atom provides one electron. Dative bond: both electrons come from one atom. NH ₃ donates its lone pair to H ⁺ .

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